MILITARY SPECIFICATION

PROJECTILE, 155MM, ILLUMINATING, M485A2
ACCESSORIES FOR, AND ASSEMBLING AND PACKING

This specification is approved for use by U.S. Army Armament Munitions, and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the component accessories and the assembling and packing for one type of projectile designated as Projectile, 155MM, Illuminating, M485A2.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

FEDERAL

FED-STD-101	-	Preservation, Packaging and Packing Materials, Test Procedures
FED-STD-151	-	Metals; Test Methods

FSC 1320

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-286	-	Propellants, Solid; Sampling, Examination and Testing
MIL-STD-414	-	Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-1169	-	Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1234	-	Pyrotechnics: Sampling, Inspection and Testing
MIL-STD-1235	-	Single and Multi-Level Continuous Sampling Procedures and Tables for Inspection by Attributes

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS (See 6.14)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

PRODUCT AND PACKING DRAWINGS

9214150	-	Projectile, 155MM, Illuminating M485A2 Loading Assembly
7549275	-	Pallet for 155MM Projectile
8796522	-	Marking Diagram and Sealing for Wood Packing Boxes
9213716	-	Candle, Loading Assembly
9213758	-	Canister, Loading Assembly

INSPECTION EQUIPMENT DRAWINGS

9201136	-	Tunnel, Light
9201268	-	Procedure, Light Output Measurement
9201390	-	Procedure, Photocell Checkout
9247071	-	Photocell
9200964	-	Photometer Head with Cell

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM E-18 - Standard Methods of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to: American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

CODE OF FEDERAL REGULATIONS

TITLE 49 - Transportation, Parts 100-199

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders should specify, "49 CFR 100-199 (latest revision)").

2.1.3 Order of precedence. In the event of a conflict between the text of this specifcation and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Projectile. The projectile assembly and parts thereof shall comply with all requirements specified on Drawings (Dwgs.) 9214150 and 9213758 and associated drawings, and with all requirements specified in applicable specifications and standards.

3.3 Moisture content.

3.3.1 First fire composition. The moisture content of the first fire composition at the loading station, at the time of loading, shall not exceed 0.30 percent, when determined as specified in 4.5.6.1 or 4.5.6.3.

3.3.2 Illuminant composition. The moisture content of the illuminant composition at the loading station, at the time of loading, shall not exceed 0.30 percent, when determined as specified in 4.5.6.2 or 4.5.6.3.

3.4 Luminosity and burning time. The average luminosity of the illuminant candle assemblies shall be not less than 1,000,000 candlepower. The average static burning time of the illuminant candle assemblies shall be not less than 110 seconds. Determination shall be as specified in 4.5.1.

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3.5 Yield strength of main parachute support. The minimum yield strength shall be as specified in the applicable ASTM referenced on dwg. 9213764 except that in the case of alloy 6070-T6, the yield strength shall be 35,000 psi minimum, measured at 0.2 percent offset. Testing shall be in accordance with 4.5.2.

3.6 Peel test of spot welded fin and canister. All spot weld buttons shall remain on the canister and each button diameter shall comply with the applicable requirement on dwg. 9213709 when tested as specified in 4.5.3.

3.7 Delay time. The delay time shall be from 6.5 seconds (min.) to 9.0 seconds (max.) for the delay holder loading assembly when tested as specified in 4.5.4.

3.8 Proving ground. The projectile shall comply with the following requirements when tested as specified in 4.5.8.

3.8.1 Functioning. The projectile shall function satisfactorily (see 6.8) and shall provide effective illumination (see 6.9) for a minimum of 100 seconds.

3.8.2 Premature bursts. There shall be no premature burst in the bore of the weapon or in the first 100 yards of flight.

3.8.3 Safety. There shall be no premature burst or evidence thereof, and there shall be no separation of metal parts either in bore or in flight (prior to the initiation of the time fuze) when fired with M119A2 propellant adjusted to give 109-117 percent of peak pressure. This requirement shall also apply to rounds fired in the safety phase.

3.9 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.10 Workmanship.

3.10.1 All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust, and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous; however, a few light scratches not exposing base material and protective coating missing per 4.4.2.32 are permitted. All required marking and stamping shall be neat and sharply defined.

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3.10.2 Painting and marking. The painting and marking shall be restored by the assembling contractor on any component furnished to him painted and marked, when such painting and markings have been marred or obliterated.

3.10.3 Rotating band. Care shall be exercised at all times so that the rotating band of the projectile is not nicked, burred or otherwise damaged.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection.

b. Quality Conformance Inspection.

4.3 First article inspection.

4.3.1 Submission of parts and assemblies. The contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantites as indicated:

PART DESCRIPTION	DRAWINGS	QUANTITY
Projectile, 155MM, Illuminating M485A2	9214150	47
Shell, Illuminant, Container	9213702	10
Delay Holder	9213708	10
Canister Assembly	9213709	10
Canister	9213710	10
Fin	9213711	10
Nose, Canister	9213712	10
Disc, Filling	9213717	10
Tube, Flash	9214147	10
Spacer	9214148	10
Support Parachute Drogue	9214152	10
Sleeve, Protector	9214603	10
Support, Parachute, Main	9213764	10
Parachute, Main	8862459	10

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Nose Canister (if applicable)	9234978	10
Parachute, Drogue, 12 inch	9251604	10
Base Canister Assy	9256843	10
Base Canister Assy	9256844	10
Pin, Swivel	9256845	10
Disc	9256846	10
Parachute, Drogue & Canister Loading Assembly	9251603	10
Container, Illuminant (prior to Loading Illuminant Comp.)	9213716	10
Container Illuminant	9213716	15
Delay Holder Loading Assy	9213707	10
Canister Loading Assy (prior to inserting shear pins)	9213758	10
Canister Loading Assy	9213758	15
Protector, Drogue Chute (split half)	9251605	10
Protector, Drogue Chute (top)	9251606	10
Parachute Main & Candle Loading Assy	9213766	10
First Fire Composition	9213716	4 ounces (ozs)
Illuminant Composition	9213716	4 ounces (ozs)

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I herein.

4.3.3 Rejection. See MIL-A-48078. If any parts or assemblies fail to comply with any or all of the applicable requirements, the first article sample shall be rejected. In the functioning test, the projectiles shall be fired and observed for compliance with the requirements specified in Paragraph 4.4.3.1, Table I. In addition, the first article sample shall be rejected if the ballistic sample is rejected. In the event that one or two projectiles show evidence of fuze failure, additional projectile(s) shall be fired to enable the determination of compliance with requirements to be based on a total of 30 projectiles. In the Safety test if any premature bursts occur, the initial production sample shall be rejected. The Government reserves the right to terminate its inspection and testing upon any failure in the first article sample to comply with any of the stated

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requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article sample. Until a first article sample is accepted, the contractor is in no way authorized by the Government to initiate regular production unless otherwise directed by the Contracting Officer.

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TABLE I. First Article Inspection.

CLASSIFICATION OF DEFECTS & TESTS MIL-P-48249A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 7		DRAWING NUMBER
	Projectile, 155MM, Illuminating, M485A2				SEE BELOW
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Projectile, 155MM, Illuminating, M485A2 (Dwg. 9214150)				
	Examination for defects	45		3.2	4.4.2.32
	Functioning	30		3.8.1	4.5.8.1
	Safety	15		3.8.3	4.5.8.2
	Contigency round for replace - mint or retested	2		--	6.8.1

NOTES:

DRSM 1 (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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TABLE I. First Article Inspection.

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48249A (AR)

PARAGRAPH	TITLE	SHEET 2 OF 7	DRAWING NUMBER
	Projectile, 155MM, Illuminating, M485A2		SEE BELOW
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Shell, Illuminant Container (Dwg. 9213702)				
	Examination for defects	10		3.2	4.4.2.1
	Delay, Holder (Dwg. 9213708)				
	Examination for defects	10		3.2	4.4.2.2
	Canister Assembly (Dwg. 9213709)				
	Examination for defects	10		3.2	4.4.2.3
	Pull test	8		3.6	4.4.3.7/4.5.3
	Canister (Dwg. 9213710)				
	Examination for defects	10		3.2	4.4.2.4
	Hardness test	8		3.2	4.4.3.8/4.5.7
	Fin (Dwg. 9213711)				
	Examination for defects	10		3.2	4.4.2.5

NOTES:

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TABLE I. First Article Inspection.

CLASSIFICATION OF DEFECTS & TESTS MIL-P-48249A (AR)

PARAGRAPH	TITLE	SHEET 3 OF 7			DRAWING NUMBER
	Projectile, 155MM, Illuminating, M485A2				SEE BELOW
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
10	Nose Canister (Dwg. 9213712) Examination for defects	10		3.2	4.4.2.6
	Disc, Filling (Dwg. 9213717) Examination for defects	10		3.2	4.4.2.7
	Tube Flash (Dwg. 9214147) Examination for defects	10		3.2	4.4.2.8
	Spacer (Dwg. 9214148) Examination for defects	10		3.2	4.4.2.9
	Support, Parachute Drogue (Dwg. 9214152) Examination for defects	10		3.2	4.4.2.10
	Sleeve, Protector (Dwg. 9214603) Examination for defects	10		3.2	4.4.2.11

NOTES:

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TABLE I. First Article Inspection.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 4 OF 7		DRAWING NUMBER
	Projectile, 155MM, Illuminating, M485A2				SEE BELOW
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Support, Parachute Main (Dwg. 9213764)				
	Examination for defects	10		3.2	4.4.2.12
	Yield strength	8		3.5	4.4.3.6/4.5.2
	Parachute, Main (Dwg. 8862459)				
	Examination for defects	10		3.2	4.4.2.13
	Parachute, Drogue, 12 Inch (Dwg. 9251604)				
	Examination for defects	10		3.2	4.4.2.14
	Nose Canister (if applicable) (Dwg. 9234978)				
	Examination for defects	10		3.2	4.4.2.15
	Base Canister Assembly (Dwg. 9256843)				
	Examination for defects	10		3.2	4.4.2.16
	Base Canister (Dwg. 9256844)				
	Examination for defects	10		3.2	4.4.2.17

NOTES:

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TABLE I. First Article Inspection.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 5 OF 7		DRAWING NUMBER
	Projectile, 155MM, Illuminating, M485A2				SEE BELOW
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Pin, Swivel				
	(Dwg. 9256845)				
	Examination for defects	10		3.2	4.4.2.18
	Disc				
	(Dwg. 9256846)				
	Examination for defects	10		3.2	4.4.2.19
	Parachute, Drogue & Canister Loading Assy				
	(Dwg. 9251603)				
	Examination for defects	10		3.2	4.4.2.20
	Pull test	5		3.2	4.4.3.3/4.5.5
	Container Illuminant (prior to loading Illumination Composition)				
	(Dwg. 9213716)				
	Examination for defects	10		3.2	4.4.2.21
	Container Illuminant				
	(Dwg. 9213716)				
	Examination for defects	15		3.2	4.4.2.22
	First fire composition	4 oz		3.3.1	4.4.3.5.1/4.5.6.1
	Illuminant composition test	4 oz		3.3.2	4.4.3.5.2/4.5.6.2
	Static functioning	5 sets		3.4	4.4.3.4/4.5.1

NOTES:

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TABLE I. First Article Inspection.

CLASSIFICATION OF DEFECTS & TESTS MIL-P-48249A (AR)

PARAGRAPH	TITLE		SHEET 6 OF 7		DRAWING NUMBER
	Projectile, 155MM, Illuminating, M485A2				SEE BELOW
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Delay Holder Loading Assy (Dwg. 9213707)				
	Examination for defects	10		3.2	4.4.2.24
	Delay time test	5		3.7	4.4.3.2/4.5.4
	Canister Loading Assy (prior to inserting shear pins) (Dwg. 9213758)				
	Examination for defects	10		3.2	4.4.2.25
	Torque test of delay assy	5		3.2	
	Canister Loading Assy (Dwg. 9213758)				
	Examination for defects	10		3.2	4.4.2.26
	Protector, Drogue Chute (split half) (Dwg. 9251605)				
	Examination for defects	10		3.2	4.4.2.27
	Protector Drogue Chute (top) (Dwg. 9251606)				
	Examination for defects	10		3.2	4.4.2.28

NOTES:

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TABLE I. First Article Inspection.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 7 OF 7			DRAWING NUMBER
	Projectile, 155MM, Illuminating, M485A2				SEE BELOW
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Parachute, Main and Candle Loading Assy (Dwg. 9213766)				
	Examination for defects	10		3.2	4.4.2.29
	Pull test	5		3.2	4.4.3.3/4.5.5
	Projectile, Prior to Assembling Lifting Plug (Dwg. 9214150)				
	Examination for defects	10		3.2	4.4.2.31

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition inspection lots of projectiles shall contain:

a. Projectile metal parts assemblies of one lot interfix number from one manufacturer.

b. Primary expelling charge assemblies from one lot interfix number from one manufacturer.

c. Secondary expelling charge assemblies of one lot interfix number from one manufacturer.

d. Main parachute assemblies from one lot interfix number from one manufacturer.

e. Drogue parachute assemblies of one lot interfix number from one manufacturer.

f. Candle loading assemblies of one lot interfix number from one manufacturer.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II. Continuous sampling in accordance with MIL-STD-1235 may be utilized if approved by the Procuring Activity.

b. Automatic gage. Facilities equipped with Government furnished automatic gaging capable of performing 100% inspection of any characteristic shall utilize this capability in accordance with the contract.

c. Alternate inspection procedures. The contractor may propose the use of alternate inspection procedures in lieu of conducting sampling inspection for defects listed in this specification (see 6.16).

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48249A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Shell, Illuminant, Container		9213702
			NEXT HIGHER ASSEMBLY
			9213716

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:	None defined				
MINOR:					
201	Outside diameter		0.65%	3.2	Gage
202	Length		0.65%	3.2	Gage
203	Thickness		0.65%	3.2	Gage
204	Perpendicularity of base with inside diameter		0.65%	3.2	Visual
205	Protective finish improper		0.65%	3.2	Visual
206	Radii missing or incorrect				
207	Concentricity of hole with inside diameter		0.65%	3.2	Gage
208	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 9213708
4.4.2.2	Delay Holder				NEXT HIGHER ASSEMBLY 9213707
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:					
101	Pitch diameter of thread, min.		0.40%	3.2	Gage
102	Pitch diameter of thread, max.		0.40%	3.2	Gage
103	Major diameter of thread, min.		0.40%	3.2	Gage
104	Flash hole missing or obstructed (2 places)		0.40%	3.2	Visual
105	Perpendicularity of rear shoulder with pitch diameter of thread		0.40%	3.2	Gage
MINOR:					
201	Concentricity of pitch diameter with small outside diameter		0.65%	3.2	Gage
202	Concentricity of outside diameter with small outside diameter		0.65%	3.2	Gage
203	Small outside diameter		0.65%	3.2	Gage
204	Diameter of delay charge cavity		0.65%	3.2	Gage
205	Length of thread		0.65%	3.2	Gage
206	Width of shoulder		0.65%	3.2	Gage
207	Distance from front face to shoulder		0.65%	3.2	Gage
208	Total length		0.65%	3.2	Gage
209	Depth of delay charge cavity		0.65%	3.2	Gage

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48249A (AR)

PARAGRAPH	TITLE	SHEET 2 OF 2	DRAWING NUMBER
4.4.2.2	Delay Holder		9213708
			NEXT HIGHER ASSEMBLY 9213707

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
210	Diameter of wrench hole (2 places)		0.65%	3.2	Gage
211	Depth of wrench hole (2 places)		0.65%	3.2	Gage
212	Position of delay holes		0.65%	3.2	Gage
213	Protective finish improper		0.65%	3.2	Visual
214	Radii, chamfers, missing or incorrect		0.65%	3.2	Visual
215	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48249A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9213709
4.4.2.3	Canister Assembly				NEXT HIGHER ASSEMBLY 9213758
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:					
101	Any component missing or assembled improperly		0.40%	3.2	Visual
102	Groove pins flush or below (4 places)		0.40%	3.2	Gage
103	Groove pins at or above min. depth (4 places)		0.40%	3.2	Gage
104	Positions of fins (3 places)		0.40%	3.2	Gage
105	Peel test	*		3.2	4.4.3.7/4.5.3
MINOR:					
201	Total length		0.65%	3.2	Gage
202	Inside diameter, min.		0.65%	3.2	Gage
203	Diameter and depth of drilled hole (4 places)		0.65%	3.2	Gage
204	Center of drilled hole to rear shoulder of nose canister (4 places)		0.65%	3.2	Gage
205	Equal spacing of welded buttons		0.65%	3.2	Gage
206	Protective finish improper		0.65%	3.2	Visual
207	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES: * Sample size as specified in applicable test method paragraph.

DRSMC-QA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9213710
4.4.2.4	Canister				NEXT HIGHER ASSEMBLY 9213709
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:	None defined				
MINOR:					
201	Total length		0.65%	3.2	Gage
202	Outside diameter		0.65%	3.2	Gage
203	Inside diameter		0.65%	3.2	Gage
204	Concentricity of inside diameter with outside diameter		0.65%	3.2	Gage
205	Distance from rear face to shoulder		0.65%	3.2	Gage
206	Distance from front face to shoulder		0.65%	3.2	Gage
207	Straightness of inside		0.65%	3.2	Gage
208	Angle missing or improper		0.65%	3.2	Visual
209	Protective finish improper		0.65%	3.2	Visual
210	Hardness test	*		3.2	4.4.3.8/4.5.7
211	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES: * Sample size as specified in applicable test method paragraph.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9213711
4.4.2.5	Fin				NEXT HIGHER ASSEMBLY 9213709
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:	None defined				
MINOR:					
201	Length of long side		0.65%	3.2	Gage
202	Thickness		0.65%	3.2	Gage
203	Profile of fin		0.65%	3.2	Gage
204	Length of flat when applicable		0.65%	3.2	Gage
205	Angle of sides		0.65%	3.2	Gage
206	Presence of 64 thru holes		0.65%	3.2	Visual
207	Spacing between horizontal holes (5 rows)		0.65%	3.2	Gage
208	Spacing between vertical holes		0.65%	3.2	Gage
209	Size of drilled holes (64 places)		0.65%	3.2	Gage
210	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER 9213712
4.4.2.6	Nose, Canister		NEXT HIGHER ASSEMBLY 9213709

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:					
101	Pitch diameter of thread, min.		0.40%	3.2	Gage
102	Pitch diameter of thread, max.		0.40%	3.2	Gage
103	Minor diameter of thread, max.		0.40%	3.2	Gage
MINOR:					
201	Outside diameter		0.65%	3.2	Gage
202	Diameter of expelling charge cavity		0.65%	3.2	Gage
203	Total length		0.65%	3.2	Gage
204	Distance from front face to shoulder		0.65%	3.2	Gage
205	Small outside diameter		0.65%	3.2	Gage
206	Concentricity of expelling charge cavity with outside diameter		0.65%	3.2	Gage
207	Concentricity of small outside diameter with outside diameter		0.65%	3.2	Gage
208	Protective finish improper		0.65%	3.2	Visual
209	Radii or chamfer missing or incorrect		0.65%	3.2	Visual
210	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9213717
4.4.2.7	Disc, Filler				NEXT HIGHER ASSEMBLY 9213716
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:	None defined				
MINOR:					
201	Outside diameter		0.65%	3.2	Gage
202	Diameter of hole		0.65%	3.2	Gage
203	Thickness		0.65%	3.2	Gage
204	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

DRSMC-QA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.8	Tube, Flash		9214147
			NEXT HIGHER ASSEMBLY 9214150

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:	None defined				
MINOR:					
201	Inside diameter		0.65%	3.2	Gage
202	Outside diameter		0.65%	3.2	Gage
203	Total length		0.65%	3.2	Gage
204	Perpendicularity of outside surface with front face		0.65%	3.2	Gage
205	Protective finish improper		0.65%	3.2	Gage
206	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9214148
4.4.2.9	Spacer				NEXT HIGHER ASSEMBLY 9214150
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:	None defined				
MINOR:					
201	Outside diameter		0.65%	3.2	Gage
202	Inside diameter		0.65%	3.2	Gage
203	Length		0.65%	3.2	Gage
204	Improperly wound		0.65%	3.2	Visual
205	Layers not glued properly		0.65%	3.2	Visual
206	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9214152
4.4.2.10	Support, Parachute, Drogue				NEXT HIGHER ASSEMBLY 9214150
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:	None defined				
MINOR:					
201	Radius		0.65%	3.2	Gage
202	Wall thickness		0.65%	3.2	Gage
203	Perpendicularity of front face with outside radius		0.65%	3.2	Gage
204	Perpendicularity of rear face with outside radius		0.65%	3.2	Visual
205	Width		0.65%	3.2	Visual
206	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

DRSM[illegible]A (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9214603
4.4.2.11	Sleeve, Protector				NEXT HIGHER ASSEMBLY 9213758
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:	None defined				
MINOR:					
201	Outside diameter		0.65%	3.2	Gage
202	Inside diameter		0.65%	3.2	Gage
203	Width		0.65%	3.2	Gage
204	Improperly wound		0.65%	3.2	Visual
205	Layers not glued properly		0.65%	3.2	Visual
206	Evidence of poor workmanship		0.65%	3.10	Visual
NOTES:					

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER 9213764
4.4.2.12	Support, Parachute, Main		NEXT HIGHER ASSEMBLY 9213758

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:					
101	Yield strength	*		3.2	4.4.3.6/4.5.2
MINOR:					
201	Radius		0.65%	3.2	Gage
202	Wall thickness		0.65%	3.2	Gage
203	Length		0.65%	3.2	Gage
204	Perpendicularity of front face with outside surface		0.65%	3.2	Gage
205	Perpendicularity of rear face with outside surface		0.65%	3.2	Gage
206	Parallelism of cut surfaces		0.65%	3.2	Visual
207	Radius missing or improper		0.65%	3.2	Visual
208	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES: * Sample size as specified in applicable test method paragraph.

DRSMC 1 (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8862459
4.4.2.13	Parachute, Main				NEXT HIGHER ASSEMBLY 9213758
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:					
101	Any component missing or improperly assembled (including stitching)		0.40%	3.2	Visual
102	Hole or tear in canopy		0.40%	3.10	Visual
103	Diameter of oval sleeve after crimping, max.		0.40%	3.2	Gage
MINOR:					
201	Overlap of tape		0.65%	3.2	Gage
202	Size of loop		0.65%	3.2	Gage
203	Color of cloth		0.65%	3.2	Visual
204	Stitching missing or broken		0.65%	3.2	Visual
205	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9251604
4.4.2.14	Parachute, Drogue 12 Inch				NEXT HIGHER ASSEMBLY 9214150
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:					
101	Length of shroud line from hem to glass cloth tape		0.40%	3.2	Gage
102	Length of glass cloth tape line from shroud line to end of loop		0.40%	3.2	Gage
103	Any component missing or improperly assembled (including stitching)		0.40%	3.2	Visual
104	Hole or tear in canopy		0.40%	3.10	Visual
MINOR:					
201	Stitching missing or broken		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9234978
4.4.2.15	Nose, Canister (if applicable)				NEXT HIGHER ASSEMBLY 9213709
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:					
101	Pitch diameter of thread, max.		0.40%	3.2	Gage
102	Pitch diameter of thread, min.		0.40%	3.2	Gage
103	Outside diameter of flange		0.40%	3.2	Gage
104	Concentricity of pitch diameter with outside diameter of flange		0.40%	3.2	Gage
105	Concentricity of inside diameter of flange with outside diameter of flange		0.40%	3.2	Gage
106	Evidence of poor weld		0.40%	3.2	Visual
MINOR:					
201	Depth of thread		0.65%	3.2	Gage
202	Thickness of flange		0.65%	3.2	Gage
203	Overall length		0.65%	3.2	Gage
204	Protective finish improper		0.65%	3.2	Visual
205	Diameter of outside hub		0.65%	3.2	Gage
206	Width of weld		0.65%	3.2	Visual
207	Minor diameter of thread, max.		0.65%	3.2	Gage
208	Radii or chamfer missing or incorrect		0.65%	3.2	Visual
209	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.16	Base Canister Assembly		9256843
			NEXT HIGHER ASSEMBLY 9213758

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:					
101	Any component missing or improperly assembled		0.40%	3.2	Visual
102	Swivel rotation		0.40%	3.2	Manual
MINOR:					
201	Crimp improper 360° over disc		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9256844
4.4.2.17	Base Canister				NEXT HIGHER ASSEMBLY 9256843
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:	None defined				
MINOR:					
201	Outside diameter		0.65%	3.2	Gage
202	Small outside diameter		0.65%	3.2	Gage
203	Hole diameter		0.65%	3.2	Gage
204	Diameter front recess		0.65%	3.2	Gage
205	Diameter rear small counterbore		0.65%	3.2	Gage
206	Diameter rear large counterbore		0.65%	3.2	Gage
207	Width of shoulder		0.65%	3.2	Gage
208	Thickness of base		0.65%	3.2	Gage
209	Depth of rear large counterbore		0.65%	3.2	Gage
210	Depth of rear small counterbore		0.65%	3.2	Gage
211	Radius of rear small counterbore		0.65%	3.2	Gage
212	Concentricity of hole with outside diameter		0.65%	3.2	Gage
213	Concentricity of rear large counterbore with outside diameter		0.65%	3.2	Gage
214	Concentricity of rear small counterbore with outside diameter		0.65%	3.2	Gage
215	Protective finish improper		0.65%	3.2	Visual
216	Radii or chamber missing or incorrect		0.65%	3.2	Visual
217	Evidence of poor workmanship		0.65%	3.10	Visual
NOTES:					

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.18	Pin, Swivel		9256845
			NEXT HIGHER ASSEMBLY
			9256843

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:	None defined				
MINOR:					
201	Diameter of pivot		0.65%	3.2	Gage
202	Diameter of ring		0.65%	3.2	Gage
203	Width of pivot		0.65%	3.2	Gage
204	Total length		0.65%	3.2	Gage
205	Diameter of hole		0.65%	3.2	Gage
206	Position of hole on ring				
207	Concentricity of ring with diameter of pivot		0.65%	3.2	Visual
208	Protective finish improper		0.65%	3.2	Visual
209	Perpendicularity of shoulder with diameter of pivot		0.65%	3.2	Gage
210	Radii missing or incorrect		0.65%	3.2	Visual
211	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

DRS 0A (D) Form 160, 1 Aug 83 replaces edition of 1 Jan 77 which may be used until exhausted.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9256846
4.4.2.19	Disc				NEXT HIGHER ASSEMBLY 9256843
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:	None defined				
MINOR:					
201	Diameter		0.65%	3.2	Gage
202	Thickness		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.20	Parachute, Drogue and Canister Loading Assy		9251603
			NEXT HIGHER ASSEMBLY 9214150

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:					
101	Any component missing or improperly assembled	*	0.40%	3.2	Visual
102	Pull test			3.2	4.4.3.3/4.5.5
MINOR:					
201	Distance between drogue and canister		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES: * Sample size as specified in applicable test method paragraph.

DRSMC 9A (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48249A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9213716
4.4.2.21	Container Illuminant (prior to loading illuminant composition)				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:					
101	Any component missing, damaged or improperly assembled		0.40%	3.2	Visual
102	Coating on side surfaces of container missing or inadequate		0.40%	3.2	Visual
103	Coating on inside surfaces of container tacky		0.40%	3.2	Visual
MINOR:					
201	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.22	Container Illuminant		9213716
			NEXT HIGHER ASSEMBLY 9213766

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:					
101	Diameter, max.		0.40%	3.2	Gage
102	Any component, missing, damaged or improperly assembled		0.40%	3.2	Visual
103	Closing disc above flush		0.40%	3.2	Visual
104	Static functioning	*		3.2	4.4.3.4/4.5.1
105	First fire composition test	*		3.2	4.4.3.5.1/4.5.6.1
106	Illuminant composition test	*		3.2	4.4.3.5.2/4.5.6.2
MINOR:					
201	Closing disc not sealed		0.65%	3.2	Visual
202	Marking missing, misleading or unidentifiable including incorrect color		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES: * Sample size as specified in applicable test method paragraph.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9213716
4.4.2.23	Container Illuminant, Packed and Sealed				NEXT HIGHER ASSEMBLY 9213766
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:	None defined				
MINOR:					
201	Container damaged to the extent contents are exposed or liable to become exposed		0.65%	3.10	Visual
202	Marking missing, misleading or unidentifiable		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.24	Delay Holder Loading Assembly		9213707
			NEXT HIGHER ASSEMBLY 9213758

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL: 1	Composition on threads		100%	3.2	Visual
MAJOR: 101	Distance to ignition powder (2 places)	*	0.40%	3.2	Gage
102	Distance to black powder (2 places)		0.40%	3.2	Gage
103	Delay time test			3.2	4.4.3.2/4.5.4
MINOR: 201	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES: * Sample size as specified in applicable test method paragraph.

DRSM A (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48249A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.25	Canister, Loading Assembly (Prior to inserting shear pins)				9213758
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 9214150 / PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:					
101	Torque test failure of delay assembly		0.40%	3.2	Test
102	Any component missing or improperly assembled (including fold of parachute)		0.40%	3.2	Visual
MINOR:					
201	Diameter of shear pin holes (4 places)		0.65%	3.2	Gage
202	Depth of shear pin holes (4 places)		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9213758
4.4.2.26	Canister, Loading Assembly				NEXT HIGHER ASSEMBLY 9214150
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:					
101	Outside diameter		0.40%	3.2	Gage
102	Total length		0.40%	3.2	Gage
103	Shear pins above flush (4 places)		0.40%	3.2	Gage
MINOR:					
201	Position of shear pins		0.65%	3.2	Gage
202	Protective finish damaged		0.65%	3.10	Visual
203	Marking missing, misleading or unidentifiable		0.65%	3.2	Visual
204	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9251605
4.4.2.27	Protector, Drogue Chute (Split half)				NEXT HIGHER ASSEMBLY 9251603
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:	None defined				
MINOR:					
201	Outside radius		0.65%	3.2	Gage
202	Inside radius		0.65%	3.2	Gage
203	Radius of inside hole		0.65%	3.2	Gage
204	Radius of teeth		0.65%	3.2	Gage
205	Length of protector		0.65%	3.2	Gage
206	No teeth in designated area		0.65%	3.2	Gage
207	Width of protrusion		0.65%	3.2	Gage
208	Location of protrusion		0.65%	3.2	Gage
209	Radii or chamfers missing or incorrect		0.65%	3.2	Visual
210	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.28	Protector, Drogue Chute (top)		9251606
			NEXT HIGHER ASSEMBLY 9251603

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:	None defined				
MINOR:					
201	Inside diameter		0.65%	3.2	Gage
202	Outside radius		0.65%	3.2	Gage
203	Outside radius to tooth root		0.65%	3.2	Gage
204	Length of protector		0.65%	3.2	Gage
205	Thickness at base		0.65%	3.2	Visual
206	Radii missing or incorrect		0.65%	3.10	Visual
207	Evidence of poor workmanship				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9213766
4.4.2.29	Parachute, Main & Candle Loading Assembly				NEXT HIGHER ASSEMBLY 9213758
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:					
101	Length of wire connecting parachute and candle loading assembly		0.40%	3.2	Gage
102	Length of loop of wire (max.)		0.40%	3.2	Gage
103	Width of loop wire		0.40%	3.2	Gage
104	Pull test	*		3.2	4.4.3.3/4.5.5
105	Any part missing or improperly assembled (including folding of main chute)		0.40%	3.2	Visual
MINOR:					
201	Diameter over crimp on sleeve, max.		0.65%	3.2	Gage
202	Diameter over crimp on first alternate sleeve, max. (if applicable)		0.65%	3.2	Gage
203	Diameter over full length cylindical crimp on either sleeve, max. (if applicable)		0.65%	3.2	Gage
204	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

* Sample size as specified in applicable test method paragraph.

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.30	Projectile (Prior to assembling base)		9214150
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR: 101	Any component missing or improperly assembled (including compound on threads and folding of parachute)		0.40%	3.2	Visual
MINOR: 202	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.31	Projectile (Prior to assembling of lifting plug)				9214150
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:					
101	Nose thread damaged		0.40%	3.2	Visual
102	Threads on lifting plug damaged		0.40%	3.10	Visual
MINOR:					
202	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9214150
4.4.2.32	Projectile				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:					
101	Position of shear pins (3 places)		0.40%	3.2	Gage
102	Diameter of bourrelet, max. (if retouched)		0.40%	3.2	Gage
103	Gap between base plug and base of projectile body)		0.40%	3.2	Gage
104	Grommet missing		0.40%	3.2	Visual
105	Lifting plug must not move upon re-application of specified assembly torque		0.40%	3.2	Test
MINOR:					
201	Grommet loose, damaged or improperly assembled		0.65%	3.10	Visual/Manual
202	Protective coating damaged (total base area in excess of 1/4 square inch)		0.65%	3.2	Visual
203	Marking missing, incorrect or unidentifiable		0.65%	3.2	Visual
204	Groove (shear) pins above flush (3 places)		0.65%	3.2	Visual
205	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.33	Pallet (Prior to strapping)				7549275
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR: 101	Cover assembly missing		0.40%	3.2	Visual
MINOR: 201	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.34	Pallet		7549275
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:					
101	Strapping loose or improperly assembled		0.40%	3.2	Visual/Manual
102	Board broken or split		0.40%	3.2	Visual
103	DoD symbol missing or incorrect		0.40%	3.2	Visual
104	Lifting plug missing or damaged		0.40%	3.2	Visual
MINOR:					
201	Marking missing, incorrect or unidentifiable		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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4.4.3 Testing. Testing is required as described herein and in the First Article and Quality Conformance Inspection Table.

4.4.3.1 Proving ground functional tests (major defect). See 3.6.

4.4.3.1.1 Production sampling. Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements specified herein, 37 sample projectiles randomly selected from each lot shall be submitted for Proving Ground Functioning Tests. The projectiles shall be fired and observed for compliance with the requirements specified in Table I below. In the event that one or two projectiles show evidence of fuze failure, additional projectile(s) shall be fired to enable the determination of compliance with requirements to be based on a total of 35 projectiles.

4.4.3.1.2 Reduced production sampling. After three consecutive lots have met the criteria of of 4.4.3.1.1, 32 projectiles shall be randomly selected for this test. Thirty (30) of the projectiles shall be fired and observed for compliance with the requirements specified in Table I below. In the event that one or two projectiles show evidence of fuze failure, additional projectile(s) shall be fired to enable the determination of compliance with requirements to be based on a total of 30 projectiles. In the event that two successive lots fail to meet the requirements of 4.4.3.1.2, the sampling provisions of 4.4.3.1.1 shall again apply. When three consecutive lots have again successfully passed the requirements contained therein the provisions of 4.4.3.1.2 shall be reinstated.

TABLE I

Category	Defect	Sample Size 35 or 30 Acc	Rej
Critical	Premature burst or metal parts separation	0	1
Major	Non-Separation with fuze functioning occurring	0	1
Major	Projectile does not provide "effective illumination" for 75 seconds or more.	1	2

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Category	Defect	Sample Size Acc	35 or 30 Rej
Major	Candle & parachute assembly does not provide for minimum "effective illumination"	SEE NOTES 1 & 2	

NOTE 1: The lot shall be rejected if the percent defective for "effective illumination" as computed with the use of MIL-STD-414 Section B Table B-3, AQL level 1.5% exceeds the maximum allowable percent defective.

NOTE 2: Only those rounds exhibiting "effective illumination" time of 75 seconds or more shall be considered in computing the percent defective per MIL-STD-414.

4.4.3.1.3 Refiring. If for any reason the proof director considers that the test conditions have detrimentally affected the test results, he shall declare the test invalid and perform a new test on additional sample rounds as required.

4.4.3.2 Delay time (see 3.7). Ten delay holder loading assemblies from each machine from each eight hours of production shall be randomly selected for this test. If any sample fails to comply with the applicable requirement, the quantity represented by the sample shall be rejected. Test equipment shall be in accordance with 4.4.4.

4.4.3.3 Pull test for crimp of oval sleeve (see dwgs. 9251603 and 9213766). One assembly shall be selected at the start of each 4 hours of production from each crimping machine to qualify the machine. A sample shall also be selected immediately following any adjustment of each machine which affects the crimping load regardless of when the adjustment is made. The test shall be performed as specified in 4.5.5 and any failure to comply with the requirements shall be cause for rejection of any production made on the applicable crimping machine until it is requalified. All sleeves shall be tested on each assembly.

4.4.3.4 Static functioning (see 3.4). The test shall be performed as specified in 4.5.1. A sample of two illuminant candle assemblies shall be selected in accordance with the sampling plan shown in 4.4.3.4.1. One candle assembly shall be tested for static burning time and candlepower. If the burning time is less than 103 seconds or the candlepower is less than 875,000, the second sample

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shall also be tested. If the average of the combined sample is less than 105 seconds or the average candlepower for the combined samples is less than 900,000, the inspection lots represented by the samples shall be rejected.

4.4.3.4.1 Sampling plan. An illuminant inspection lot shall be as defined in paragraph 6.6.

4.4.3.4.1.1 A sample of two candles from each inspection lot shall be selected. The canister assembly shall be tested for compliance with the requirements specified in 4.4.3.4. This sampling frequency shall be continued until 15 consecutive samples (15 groups of 2 candles) comply with the specified requirements.

4.4.3.4.1.2 After successful completion of 4.4.3.4.1.1, the sampling frequency may be reduced to two candles from each group of six inspection lots (approximately 510 candles). The canister assembly shall be tested for compliance with the requirements specified in 4.4.3.4. This sampling frequency shall be continued until 15 consecutive samples (15 groups of 2 candles) comply with the specified requirements. In the event that one lot fails to comply with the requirements, the sampling shall revert to 4.4.3.4.1.1.

4.4.3.4.1.3 After successful completion of 4.4.3.4.1.2, the sampling frequency may be reduced to two candles from each group of fifteen inspection lots (approximately 1275 candles). The canister assembly shall be observed for compliance with the requirements specified in 4.4.3.4. This sampling frequency shall be continued until 15 consecutive samples (15 groups of 2 candles) comply with the specified requirements. In the event that one lot fails to comply with the requirements, the sampling shall revert to 4.4.3.4.1.2.

4.4.3.4.1.4 After successful completion of 4.4.3.4.1.3, the sampling frequency may be reduced to two candles from each eight hours of production (approximately 1615 candles). The canister assembly shall be tested for compliance with the requirements specified in 4.4.3.4. In the event that one lot fails to comply with the requirements, the sampling shall revert to 4.4.3.4.1.3.

4.4.3.4.1.5 In the event of any production stoppage greater than one week, the sampling plan shall revert to 4.4.3.4.1.2 and in the event of production stoppage greater than two weeks, the sampling plan shall revert to 4.4.3.4.1.1. If the mixing operation has not been shut down, but merely diverted to another item, the sampling plan shall revert to 4.4.3.4.1.2.

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4.4.3.4.1.6 In the event of failure of any sample from a multiple inspection lot to comply with the requirements, samples from each individual inspection lot shall be tested for compliance with the requirements of paragraph 4.4.3.4. Failure of an inspection lot to comply with the requirements shall result in rejection of that particular inspection lot.

4.4.3.5 Moisture content.

4.4.3.5.1 First fire composition. The contractor shall provide adequate controls to insure compliance with the requirements and shall select a three to four ounce first fire sample from each eight hours of production and place it in an air tight labeled container indicating the name of the material and date of production. If the sample after analysis fails to comply with the applicable requirements, the eight hours of production represented by the sample shall be rejected.

4.4.3.5.2 Illuminant composition. The contractor shall provide adequate controls to insure compliance with the requirements and shall select a three to four ounce illuminant sample from each eight hours of production and place it in an air tight labeled container indicating the name of the material and date of production. If the sample, after analysis, fails to comply with the applicable requirements, the eight hours of production represented by the sample shall be rejected.

4.4.3.6 Yield strength of main parachute support (see dwg. 9213764). Out of each lot of main parachute support produced, five (5) sample items shall be randomly selected for yield strength testing. If any sample (test specimen) fails to comply with the applicable requirement, the lot shall be rejected. The test shall be performed as specified in 4.5.2.

4.4.3.7 Peel test of welded fin and canister (see dwg. 9213709) and paragraph 3.6). Two sample canister metal parts assemblies shall be selected at random from each four hours of production from each welding machine. One fin from one of the sample canisters shall be tested as specified in 4.5.3. If two or more weld buttons fail to comply with the requirements specified, the four hours of production from the welding machine represented shall be rejected. If one button fails the requirements specified, a welded fin from the second sample canister shall be subjected to the test. If in the second sample one or more weld buttons fail to meet the requirement specified, the four hours of production from the welding machine shall be rejected.

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4.4.3.8 Hardness test of canister body (see dwg. 9213710). Samples shall be randomly selected from each eight hours of production in accordance with MIL-STD-105, using Table II A, Code letter D and AQL 1.5 percent. If any of the hardness determinations taken on the sample canister bodies fail to comply with the requirements specified, the production represented by the sample shall be rejected. The test shall be performed in accordance with 4.5.7. In the event that a continuous heat treating process is not being used, the sample shall be selected from each heat treat batch instead of each eight hours of production.

4.4.4 Inspection equipment. For inspection equipment required to perform the examinations and tests prescribed herein the contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.4.4.1 Automatic inspection equipment. All automatic test and inspection equipment when utilized by the contractor shall be subjected to periodic verification and/or calibration check during production. The procedures shall include as a minimum: Frequency; method of verification; and the method of retrieval of items produced between verification/calibration. These procedures shall be included in the operation/calibration procedures as required by the above paragraph.

4.5 Test methods and procedures.

4.5.1 Static functioning test. The test equipment required for this test is delineated in paragraph 2.1 under Inspection Equipment Drawings. The temperature of the candle shall be $70^{\circ} \pm 5^{\circ}$ at time of test. In calculating the candlepower of any one illuminant assembly, the best continuous one hundred seconds of illuminometer readings are to be used. The first fire disc assemblies shall be removed prior to testing. The distance from the photometric cell to the flare shell shall be 80.00 ± .25 feet. The axial center of the photometric cell and the axial center of the fireball shall be located 54.0 ± .5 inch from the floor. The temporary back wall listed on dwg. 9201136 shall be removed so that the distance from the center of the candle to the back wall is 14-15 feet. The center line of baffle opening shall be in line with axial center of fireball and axial center of photometric cell. The test shall be conducted in a hearth that has been painted flat black. Depending on ash buildup the tunnel hearth shall be repainted at least once in a three month duration to maintain the black interior or more frequently as determined by the resident Government quality assurance representative.

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4.5.2 Yield strength of main parachute support. One specimen shall be machined from the longitudinal center of each of the five items in the sample and be tested to determine compliance with the applicable requirement. Testing shall be performed in accordance with Federal Standard 151, using type R3 specimens.

4.5.3 Peel test of welded fin and canister. This test shall be performed by peeling a fin back against the welds until failure occurs around the periphery of the welds or the fin fails. Failure of the base metal outside the weld area shall be considered evidence that a weld button has formed.

4.5.4 Delay time. The delay holder loading assembly shall be assembled in a test fixture. An M17 detonator (or another suitable initiator which will assure accurate determination of the delay time of the assembly) in a fixture shall be placed upon the holder in line with one delay train. The detonator shall be initiated and the delay time shall be recorded. Rotate the delay holder loading assembly fixture 180 degrees and repeat the procedure.

4.5.5 Pull test. The test specimen shall be assembled in a universal testing machine and the load applied in an axial direction until the crimp or wire breaks. The rate of travel of the testing machine head shall be 0.100 to 0.125 inch per minute.

4.5.6 Moisture content.

4.5.6.1 First fire composition. The moisture content of the first fire composition shall be determined in accordance with Standard MIL-STD-1234, Method 101.3. The special solvent to be used shall be equal parts by volume of Dimethyl-formamide and anhydrous methanol.

4.5.6.2 Illuminant composition. The moisture content of the illuminant composition shall be determined in accordance with the Karl Fischer Method as stated in Standard MIL-STD-1234, Method 101.2 up to paragraph 5.3. A 50 ml sample of 1:1 dimethyl-formamidemethyl alcohol solution shall be added into a 250 ml Erlenmeyer flask (fitted with a rubber topper) containing 10 grams of the pyrotechnic sample.

(Exercise care to exclude atmospheric moisture). The flask shall be stoppered and shaken vigorously to thoroughly mix solution and pyrotechnic sample. Allow the solution to set 5 to 15 minutes. A blank sample of the 1:1 dimethylformamide methanol solution shall be prepared as described above. A 60 ml portion of methanol shall be transferred through the sample porthole into the dry titration beaker

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containing a plastic coated magnet. Stopper at once. The magnetic stirrer shall be started and the water back-titrated in this solvent by adding a slight excess of Karl Fischer reagent to produce a reddish-brown color which persists for at least half a minute. The solution shall be titrated slowly with the water-in-methanol solution until the exact preliminary end point is reached. All succeeding operations shall be completed within 3.5 - 0.25 minutes. 25 ml of the supernatant solution shall be rapidly introduced, accurately pipetted from the supernatant liquid into the beaker through the same porthole of the rubber stopper. The porthole shall be closed immediately and the contents of the beaker stirred for thirty seconds by means of the magnetic stirrer. A slight excess (10 - 15 mls) of the Karl Fisher reagent shall be added and carefully back-titrated with the standard water-in-methanol solution to the final and exact end point. A water determination of the dimethylformamide-methanol blank solution shall be conducted as specified above for the extraction of water from the sample. The percent water in the pyrotechnic sample shall be calculated as follows:

Calculation:

$$\frac{AB}{5(10)} \quad \text{Percent } H_2O$$

Less _____ Std. Blank

Leaves _____ Percent Moisture in Sample

Where:

A = Mls of Karl Fischer to titrate sample
B = Standardization Factor (mg/ml)
10 = Constant Factor
5 = Standard Sample Weight

4.5.6.3 Alternate method for first fire and illuminant composition moisture. The moisture content of the first fire composition and illuminant composition shall be determined in accordance with Method T101.4 Standard MIL-STD-286. The temperature setting shall be 100 degrees C and time setting 30 minutes; the sample size shall be between 0.25 and 0.50 grams.

4.5.7 Hardness test of canister body (non-destructive test). Six Rockwell hardness determinations, shall be made on each sample canister body and shall be the average of the six determinations. Each determination shall consist of the average of two readings taken approximately 1/4 inch apart. Two determinations, taken 180 degrees apart, shall be taken at each end and also two determinations shall be taken at approximately the middle of the canister body. Testing procedures shall be in accordance with Method 243.1 of Federal Standard 151 using equipment in accordance with 4.4.4. After testing, conforming samples may be returned to production.

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4.5.8 Proving ground. These tests shall be conducted at a Government proving ground in accordance with the applicable Acceptance Test Procedure. The round shall be fitted with an M565 or an M577 fuze. The sample projectile shall be fired with an M4 or M119A2 (subject to firing limitation) propelling charge from an M185 cannon. Observation shall be made for conformance to all requirements (see 6.8 and 6.9). The sample projectile shall be tested at the specified zone using weapons, equipment methods and procedures in accordance with the applicable Acceptance Test Procedure. Tubes for this test shall not have less than twenty-five (25) percent remaining tube life based on current gun tube condemination criteria.

4.5.8.1 Functioning. The sample projectile shall be fired from an M185 cannon, using the appropriate propelling charge for zone 7. The fuze setting shall be 27 seconds and the time for effective illumination shall be recorded for projectile fired (see 6.8 and 6.9).

4.5.8.2 Safety. The sample projectile shall be fired from an M185 cannon with an M119A2 propelling charge adjusted to give a pressure of 109 - 117 percent
of peak pressure at zone 7.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Packaging. Inert components shall be in accordance with the requirements of MIL-STD-1169.

5.2 Packing.

5.2.1 Domestic shipment. Packing and marking of inert components shall be in accordance with requirements of specification MIL-STD-1169, Level B.

5.2.2 Level C. Illuminant candle assemblies shall be packed in accordance with Code of Federal Regulations "49 CFR 173.91 for special fireworks.

5.2.3 Level A. The projectile shall be palletized in accordance with dwg. 7549275.

5.3 Marking.

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5.3.1 Marking. In addition to any marking required by the contract or order, marking of the illuminant candle assembly shall be in accordance with Code of Federal Regulations "49 CFR 100-199" and dwg. 8796522. DOT Marking - SPECIAL FIREWORKS - HANDLE CARFULLY - KEEP FIRE AWAY.

5.3.2 Marking. Marking of the projectile pallet shall be in accordance with dwg. 7549275.

6. NOTES

6.1 Intended use. Not applicable to this specification.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs as required to Commander, US Army Armament Research and Development Center, ATTN: DRSMC-QAR-Q (D), Dover, NJ 07801.

6.4 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9(n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph	Data Requirements	Applicable DID
6.16	Procedure Report	* DI-T-5372 (Tailored)
4.4.3.4, 4.4.3.7, 4.4.3.5.1, 4.4.3.5.2, 4.5.6.3 (if applicable)	Quality Inspection Test Demonstration & Evaluation Report	DI-R-1724
4.4.3.2, 4.4.3.3, 4.4.3.4, 4.4.3.6, 4.4.3.8	Quality Inspection Defect Report	DI-R-1721

* Delete a, b, c, d, of paragraph 1 block 10, DD Form 1664, and substitute contractor designated parameters and limits

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6.5 Submission of ballistic test data and data cards. In addition to the normal distribution, one copy of all ballistic data and ammunition cards shall be forwarded to Commander, US Army Armament Research and Development Center, ATTN: DRSMC-QAR-Q (D), Dover, NJ 07801.

6.6 Illuminant inspection lot. Illuminant from the mixer will be collected in containers of approximately equal quantity of illuminant per container which will be numbered and used in sequence. Illuminant candle assemblies will be assigned an inspection lot number for each 85 candles produced. Smaller lot sizes will be accepted when dictated by interruption or shutdown of operations.

6.7 Static burning time. Minimum static burning time as referenced herein (see 3.4) is required, as a time base for the determination of luminosity and compliance does not in itself guarantee final ballistic minimum burning time requirement. Structual integrity of the candle under set back loading is of prime importance in meeting the ballistic burning time requirement and is attained by proper formulation, blending, consolidation and cure.

6.8 Functioning. Satisfactory functioning shall be defined as ejection of the canister assembly and function of the delay assembly and ejection of the parachutes and illuminant assembly and proper suspension and ignition of the illuminant such as to assure "effective illumination" (i.e., fuze function shall be disregarded).

6.8.1 Improper functioning. Fuze failures are not considered as improper functioning. Two extra projectiles are furnished with each initial production and first production lot and two extra projectiles are furnished with each subsequent production lot. In the event of fuze failures, the extra projectiles are fired as necessary to obtain the required sample size. If, after firing the extra projectile, 25 or 30 proper fuze functions have still not been obtained as required, two alternatives exist: If the lot meets the acceptance requirements even when fuze failures are considered as improper functions, the lot is acceptable; if the lot does not meet the requirements when fuze failures are so considered, it must be referred.

6.8.2 Premature functioning. A premature function is defined as any which occurs in the bore of the weapon or in the first 100 yards of flight. Any function which occurs after 100 yards of flight but before fuze set time is arbitrarily considered as a fuze premature and is disregarded.

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6.9 Effective illumination. Effective illumination is defined as burning of the canister without exploding or breaking up while suspended by the parachute. The time recorded for "effective illumination" shall be considered as the time from the appearance of first light from the candle to either (1) burnout of the candle while still in the air, or (2) burning of the candle until ground impact. Burning of the canister after striking the ground shall not be considered as "effective illumination". Any occurrences which detract from "effective illumination" as herein defined, shall be recorded on the official firing record.

6.9.1 Mean effective illumination time. Rounds on which fuze failures occurred should not be used in calculating the mean. All other improper functioning rounds must be used in calculating the mean.

6.10 Proving ground. The proving ground shall make the necessary arrangements to recover all malfunctions which may occur during ballistic test unless otherwise directed by the procuring agency. The terrain in the area specified by the ballistic test conditions from the gun site to down range shall be suitable so that all rounds fired including malfunctions will be accessible for recovery.

6.10.1 Rounds to assemble for proving ground. The number and types of rounds required are shown in Table II.

TABLE II

Phase	Type Round	Purpose	Initial Production a	Regular Production b
Functioning	Stock	Verification of elevation & fuze setting	As required	As required
	Test	Test	35	30

NOTE:

a. Until three consecutive lots have been approved.

b. After three consecutive lots have been accepted.

c. See paragraph 6.8.1.

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6.11 Submission of first article samples. Instructions as to the location of the evaluation of the initial production sample shall be obtained from the contracting officer. Upon receipt of such request, the contracting officer shall advise US Army Armament Research and Development Center ATTN: DRSMC-QAR-Q (D), Dover, NJ 07801.

6.12 Visual examination qualification. When compliance with the applicable requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.13 Statements of findings shall show the specific analytical results of each test or inspection performed as required by the applicable drawing or specification together with the specified acceptable limits.

6.14 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Center (ARDC) may also include drawings prepared by and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDC.

6.15 Automatic inspection equipment. All automatic test and inspection equipment when utilized by the contractor shall be subjected to a periodic verification and/or calibration check at frequent intervals during normal production. The procedures shall include as a minimum: Frequency; method of verification; and the method of retrieval of items produced between verification/calibration. These procedures shall be included in the operation/calibration procedures as required by paragraph 4.4.4.

6.16 Alternate inspection procedures.

6.16.1 Justification for alternate inspection procedures. Submitted documentation must include sufficient information to indicate an economic benefit, and must also clearly identify that there is no reduction in product quality.

6.16.2 Alternate inspection procedures. The alternate inspection procedures as a minimum must include:

a. Frequency (production interval) for validating equipment functioning.

b. Proposed method of verification to include process and inspection drawings.

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c. Description of material control procedures to be imposed during the production interval between validations (Note: Under these provisions final acceptance will be withheld from material produced during the production interval until conducting satisfactory validation).

6.16.3 Approval of alternate inspection procedures. Proposed plans shall be submitted per paragraph 6.5 for approval.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-48249A	PROJECTILE, 155MM, ILLUMINATING, M485A2

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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